

ROCKET COMPETITION A4-INTERNATIONAL CATEGORY

TECHNICAL SPECIFICATIONS 2026



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VERSIONS			
Version	Date	Remarks	Changes
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1.1	17.01.2026	Revision	3.2.10.
1.2	04.02.2026	Revision	3.3.4.4., 3.3.2.14.
1.3	20.02.2026	Competition Calendar	Application Deadline
1.4	02.04.2026	Quota Adjustment	Number of Teams Advancing After Objections
1.5	03.04.2026	Quota Adjustment	A4 PDR Quota
1.6	15.04.2026	Competition Calendar	Update the process schedule

1. DEFINITIONS

- **APOGEE:** The highest altitude reached by the rocket during a flight.
- **CATEGORY:** The groups that describe the different tasks determined by the Competition Committee and the restrictions of the functions.
- **CHECKLIST:** The list created by the teams to check that they have all their materials with them in the firing test area and that all integration activities and final preparations before the launch have been carried out completely.
- **COMMERCIAL SYSTEM (COTS):** Ready-to-use off-the-shelf systems designed and manufactured by a commercial company.
- **COMMUNICATION COMPUTER:** This is the system that transmits telemetry data from the rocket to the ground station through the flight communication subsystem. It can be a standalone system or integrated with the Flight Control Computer.
- **CONTROL SURFACES:** Moving surfaces that provide the necessary aerodynamic forces/moments for the rocket to perform the targeted maneuvers.
- **EYE BOLT:** It is an eye-shaped intermediate element used within the rocket to connect the parachute and the rest of the rocket's systems.
- **FLIGHT CONTROL COMPUTER:** A subsystem in which raw data from sensors is collected, stored, and managed to facilitate flight control. The separation and recovery systems found on the rocket are managed by the flight control computer.
- **GROUND STATION:** Stations developed by teams to receive the telemetry data produced by the rocket during the flight and to track the flight.
- **IGNITION WIRE:** A wire in the burner section that ignites the fuel and/or flammable material when an electric current is passed through its open end.
- **INTEGRATION FRAME:** The parts used to connect the two frames of the rocket to each other.
- **LAUNCH RAIL:** The launch rail from which the rocket will be launched is inclined at a set angle to the ground, in accordance with the competition arrangements.
- **MECHANICAL ASSEMBLY AND FASTENERS:** The body that holds the Control Section elements together as well as the screws, nuts, pins, washers, couplings, etc. used for fastening and bearings and mechanical elements used for bedding purposes.
- **MOTOR:** The section of the rocket that provides the thrust.
- **MOTOR CLASS:** The class in which the thrust level ranges are determined according to international standards.
- **OPEN ROCKET:** An open-source used for sounding rockets allowing the design and the flight simulation of the rockets (*more information can be found at <http://openrocket.info>*)
- **OPERATIONAL CONCEPT:** The concept describes the mission environment and the tasks to be performed from the launch of the rockets to the end of the flight.
- **OUT OF COMPETITION FLIGHT:** A non-scored flight by teams that have failed to obtain the necessary permits for a flight in accordance with the competition specifications despite having obtained the necessary security labels from the referees on the competition site.

- **PAYLOAD:** The systems that must be launched to a certain altitude along with the rocket and safely landed after stage or rocket component separation. They're designed to perform specific tasks when teams use scientific payloads.
- **PRIMARY PARACHUTE:** The parachute deployed at the apogee brings the rocket's velocity down to a certain level, ensuring a controlled and drift-free decrease in altitude until the secondary parachute is deployed.
- **RAIL BUTTON:** A manual component that is mechanically attached to the rocket's frame and that ensures the rocket is positioned linearly on the launch rail. When rockets are fired, they accelerate in a linear direction on the launch rail.
- **RECOVERED PAYLOAD:** The payload that lands with a parachute after separation from the rocket during flight, on which no damage is detected in the inspections made, and which is considered suitable for a new mission.
- **RECOVERED ROCKET:** A rocket on which no damage to its components (*frame, flight computer, etc.*) is detected in the inspections made after a flight, and which is considered suitable for a new mission.
- **RISK ANALYSIS:** An analysis that the participating teams must complete assessing certain aspects of their rocket designs.
- **ROCKET:** A vehicle designed to safely launch its payload at a certain altitude, which can use different motors for different categories as stated in the document, and move under thrust at the beginning of their trajectory but subsequently travel according to ballistic laws.
- **SECONDARY PARACHUTE:** The parachute that comes out of the rocket whose speed has been reduced to a certain level by opening the primary parachute and allowing the rocket to land on the ground at a safe speed so that it can be recovered.
- **STAGED ROCKET:** Rockets with two or more engines, with each stage having its own thrust components (*series or parallel*).
- **STATIC MARGIN:** Static margin (SM) is a dimensionless number obtained by dividing the distance between the center of pressure (X_{CP}) and the center of gravity (X_{CG}) of the rocket by the rocket body diameter (D_B)
$$\frac{X_{CP} - X_{CG}}{D_B} = SM$$
- **TEKNOFEST ROCKET COMPETITION COMMITTEE:** The authorized committee, composed of representatives of the Turkish Technology Team (T3) Foundation, ROKETSAN Inc., responsible for the planning, coordination and execution of the TEKNOFEST Rocket Competition.
- **TELEMETRY DATA:** The collective of meaningful data collected from sensors on the system during flight that is transmitted to the ground station.

2. INTRODUCTION

The main purpose of the TEKNOFEST Rocket Competition, organized by ROKETSAN Inc. under the coordination of the Turkish Technology Team (T3), is to increase the interest of the youth in the field of aviation and space, to support them interested in rockets and missiles, to encourage them to conduct research into future technologies, to familiarise them with professional design processes, to raise awareness in society about the technological achievements of young people and to increase the nation's confidence in the future. Participants with knowledge and experience in different disciplines carry out work in teams (*design, production, integration, analysis/modelling, testing, etc.*), gaining technical and scientific reporting competencies. In addition, by meeting and exchanging ideas with other teams with similar interests and with professionals working in these fields, the teams develop themselves with innovative ideas.

The TEKNOFEST Rocket Competition, which was first held in 2018, was Europe's first rocket competition and the second in the world, and, as can be seen from the team application data for subsequent years, the interest of young people is constantly increasing.

YEARS	2018	2019	2020	2021	2022	2023	2024	2025
APPLICANT TEAM	156	570	516	544	455	473	1098	1175

Allowing young people to produce and launch rockets of their own design will encourage them to make career choices focused on science, technology, and engineering in the future. Using their knowledge in various fields of science and engineering, young people engage successfully in the design, production, and integration of rockets through interdisciplinary thought and the planning of complex operations/processes. Additionally, students benefit from the experiences of other successful teams and gain valuable experience by examining different design examples. The achievements of the young people participating in this competition are summarized below:

- Transforming high-level (*system level*) requirements into more detailed (*subsystem and component levels*) requirements,
- Team members working in different disciplines (*Aerodynamics, Avionics, Structural Integrity, Payload, and Recovery Systems*) create a conceptual design and then produce a detailed design following a Systems Engineering approach,
- Utilizing optimization methods and applications in the design process,
- Carrying out production and supply processes for the productization of the design,
- Creation and implementation of assembly and integration plans,
- Creating a Testing Verification Plan for the verification and validation of the resulting product and implementing the plan,
- Managing all processes from the outset (*receipt of specifications*) to the final stage (*launching the rocket*), in accordance with Project Management principles,
- Analyses and management of the risks.

The most important outcome of this competition is to encourage young people who like and care about their work, who can organize well with team spirit, who have sufficient technical knowledge and perceive engineering correctly, and who can foresee risks in sufficient time and take the necessary proactive precautions.

Applications for the TEKNOFEST Rocket Competition, which will be held for the ninth time in

2026, are to be made through the official website of the Aviation, Space and Technology Festival TEKNOFEST Technology Competitions (www.teknofest.org) before the deadline specified in the Competition Calendar. We wish success to all teams that apply to the competition.

3. REQUIREMENTS

3.1. COMPETITION CATEGORIES

3.1.1. The categories in the competition are presented in **Table 3.1** below.

GROUP	NO	MAIN MISSION	CATEGORY
A	1	Development of a Single-Stage Rocket with a COTS Solid Propellant Rocket Motor to Reach the Highest Altitude	High-School Category
	2		Medium-Altitude Category
	3		High-Altitude Category
	4		International Category
	5	Development of a Two-Stage Rocket with a COTS Solid Propellant Rocket Motor to Reach the Highest Altitude	Challenging Task Category
	6	Development of a Single-Stage Rocket with an SRAD (<i>Student Researched and Developed</i>) Hybrid Propellant Motor to Reach the Highest Altitude	SRAD Hybrid Propellant Motor & Rocket Category
	7	Development of a Single-Stage Rocket with an SRAD (<i>Student Researched and Developed</i>) Liquid Propellant Motor to Reach the Highest Altitude	SRAD Liquid Propellant Motor & Rocket Category
B	1	Development of Rocket Flight Control System	SRAD Rocket Control Section Development Category
	2	Development of Rocket Propulsion System	SRAD Hybrid Propellant Rocket Motor Development Category
	3		SRAD Liquid Propellant Rocket Motor Development Category
	4	Development of Flight Simulation, Analysis & Modeling Software for Rocket Design	Rocket Flight Modeling and Simulation Software Category

Table 3.1 Competition Categories

3.1.2. The coding system of the categories is presented in **Table 3.2** below;

CODE	KATEGORİ	CODE	KATEGORİ
A1	High-School	B1	SRAD Rocket Control Section Development
A2	Medium-Altitude	B2	SRAD Hybrid Propellant Rocket Motor Development
A3	High-Altitude	B3	SRAD Liquid Propellant Rocket Motor Development
A4	International	B4	Rocket Flight Modelling & Simulation Software
A5	Challenging Task		
A6	SRAD Hybrid Propellant Motor & Rocket		
A7	SRAD Liquid Propellant Motor & Rocket		

Table 3.2 Competition Categories

3.2. GENERAL REQUIREMENTS FOR A4 CATEGORY

3.2.1. Applications to the competition shall be submitted according to the schedule announced on the relevant page of the official TEKNOFEST website, and any applications submitted after the deadline will not be considered.

3.2.2. Participation as a team is mandatory.

3.2.3. Teams must consist of a minimum of six (6) and maximum 15 (fifteen) members.

3.2.4. Teams may consist of both students or alumni.

3.2.5. Students from different educational institutions may participate in the competition as mixed teams

3.2.6. At least half of the team members must be students.

3.2.7. The teams may bring a minimum of four (4) and a maximum of 6 (six) team members to the firing test range.

3.2.8. When evaluating all team members who participate in the assembly/integration activities at the Aksaray Hisar Firing Range, if the number of female members is equal to or greater than half of the total number of members, the team will be awarded an additional 100 points.

3.2.9. Among the teams from foreign countries applying to the A4 category, the number of citizens of the Turkish Republic (*including those who hold Turkish citizenship in their dual citizenship*) may be at most half of the total team members. For the teams applying to the A4 category, citizens of the Turkish Republic (*including those who hold Turkish citizenship in their dual citizenship*) may benefit from this provision provided that they are currently students in the country to which the team is affiliated, or are graduates of an educational institution in that country and currently reside there (*student status, graduation, and residency must be documented and officially proven to the Competition Committee*).

3.2.10. Applications to the A4 Category will not be accepted from teams whose members are all foreign nationals and are active students (associate, undergraduate, or graduate) studying at universities in Türkiye or the Turkish Republic of Northern Cyprus, or graduates residing in Türkiye.

3.2.11. The teams must design, produce, and launch rockets that will reach the highest altitude provided that the rockets exceed the minimum altitude given in **Table 3.3**. The flight performance of teams that do not meet the minimum altitude will not be considered successful;

CATEGORY	MINIMUM ALTITUDE
A4 International Category	8,000 ft

Table 3.3 Minimum Altitude for Category A4

3.2.12. All teams will compete to reach both the highest altitude (*higher than the minimum altitudes in Table 3.3*) and the target altitude officially declared to the TEKNOFEST Rocket Competition Committee prior to the launch with a maximum tolerance of $\pm 15\%$ (fifteen percent).

3.2.13. All teams must design and manufacture a rocket that fully meets all the indispensable requirements specified in the TEKNOFEST Rocket Competition Specification, shall perform ground tests at the assembly/integration of COTS/SRAD components/subsystems levels, and shall successfully launch the rocket (*in accordance with the concept of operation*) using the rocket motor to be provided by the TEKNOFEST Rocket Competition Committee (*the motors shall be handed over to the teams in the firing test range*).

3.2.14. Each team can participate in the competition with only one (1) rocket.

3.2.15. The competition schedule is presented below;

STAGE CATEGORY	Application	Preliminary Design Review (PDR)	Critical Design Review (CDR)	Launch Readiness Review (LRR) and Firing Tests
A4	T ₀	T ₀ + 2 Months	T ₀ + 4 Months	T ₀ + 7 Months

Table 3.4 Schedule for the A4 Category (*in terms of months*)

3.2.16. No member of a team competing in the A4 competition category can be part of another team competing in the same category.

3.2.17. The deadlines for the submission of the reports to be prepared by all teams are presented below:

REPORT CATEGORY	Application	Preliminary Design Review (PDR) Report	Critical Design Review (CDR) Report	Launch Readiness Review (LRR) Report
A4	T ₀	T ₀ + 9 Week	T ₀ + 17 Week	T ₀ + 31 Week

Table 3.5 Report Delivery Schedule (*in terms of weeks*)

3.2.18. The deadlines for the submission of reports are as stated in the TEKNOFEST-2026 Rocket Competition Calendar. The teams' reports that do not comply with this schedule will not be taken into consideration.

3.2.19. The teams are responsible for preparing the relevant reports (*Preliminary Design Review-PDR, Critical Design Review-CDR, and Launch Readiness Review-LRR*) in accordance with the deadlines specified in the competition calendar and the standards provided by the TEKNOFEST Rocket Competition Committee. The teams' reports that do not comply with those conditions will not be taken into consideration.

3.2.20. Teams advancing through the stages in the A4 category will be determined based on available quotas.

3.2.21. In the PDR, CDR, and LRR stages, teams will be evaluated according to a systematically determined ranking, and among the teams evaluated in this order, those who achieve the threshold score for the respective stage will be eligible to qualify within the available quota.

3.2.22. For the A4 category, the minimum score required for the teams to advance to the next stage is 85% of the total points available in the relevant stages (*PDR, CDR and LRR*).

3.2.23. In the A4 category, teams in A4 category advancing to the next stage at each stage (*PDR, CDR and LRR*) will be determined according to a threshold score (*the threshold to be set separately by the Competition Committee and while determining the threshold score, the quota for the respective stage will be taken into consideration*).

3.2.24. The quotas for A4 category based on stage are listed below;

CATEGORY	QUATOS		
	NUMBER OF TEAMS ELIGIBLE TO ADVANCE TO THE CDR STAGE AFTER THE PDR EVALUATION	NUMBER OF TEAMS ELIGIBLE TO ADVANCE TO THE LRR STAGE AFTER THE CDR EVALUATION	NUMBER OF TEAMS ELIGIBLE TO BECOME FINALISTS AFTER THE LRR EVALUATION
A4	30 + NTAO	8 + NTAO	6

*NTAO: Number of Teams Advancing After Objections

Table 3.6 Quatos of A4 Category Based on Stages

3.2.25. The threshold score required to successfully complete each stage (*PDR, CDR and LRR*) will be determined separately by the Competition Committee, and this threshold score will also be announced when the evaluation results are published, in accordance with the TEKNOFEST Rocket Competition Calendar. The threshold score in question will be determined separately by the Competition Committee for each year and each category.

3.2.26. All teams competing in the A4 category shall have two opportunities to object the Competition Committee evaluations in the respective stages (*PDR, CDR and LRR*) namely "INITIAL OBJECTION" & "HIGH-LEVEL OBJECTION", respectively.

3.2.27. All teams competing in the A4 category will have a single right to object to the scores (*the scores received from assembly/integration, launch, and recovery activities*) they obtain in the finals at the Aksaray Hisar Firing Range.

3.2.28. For the teams competing in the A4 category, the target altitudes will be the ones submitted in written form to the Competition Committee by the teams after the assembly/integration activities at the Aksaray Hisar Firing Range.

3.2.29. The teams are responsible for listing all team members and team advisors who participated in the competition in their reports.

3.2.30. Each team must participate in the competition with one (1) advisor.

3.2.31. Teachers, academicians, instructors, and members of teams that have previously won the right to launch a rocket in any rocket competition (*provided they are over 18 years of age*) can participate as advisors.

3.2.32. For teachers, academicians, and trainers who are to serve as consultants, an officially approved electronic copy of the teacher/trainer/academician ID card given to the consultant by the educational institution in which she/he works (*photocopy of the ID card given by the institution in which she/he works will have the phrase “AS THE ORIGINAL” on it, and this document will be scanned and signed by the document manager*) must be uploaded to the system by the team during the CDR phase.

3.2.33. If members of teams that have previously won the right to launch rockets in any rocket competitions are to serve as consultants, documents proving the consultant's participation in the competition (*documents provided to the consultant by the competition organization and a photograph of the consultant taken in the competition, if exists*) must be uploaded to the system by the team during the CDR phase.

3.2.34. An electronic copy of a document indicating that the persons who will serve as consultants will fulfill their consultancy duties (*the submission document must be signed by the consultant and scanned to be entered into the system*) must be uploaded to the system by the team during the CDR phase.

3.2.35. Applications from the teams that do not comply with the restrictions regarding consultant-related issues will not be taken into consideration.

3.2.36. In the event of a change of advisor, the reason for the change and the new advisor information must be immediately declared in written form to the TEKNOFEST Rocket Competition Committee. Advisor changes must be made at least 24 hours before the start date of the TEKNOFEST Rocket Competition finals, otherwise, the advisor change will not be accepted.

3.2.37. If the team ranks and is entitled to receive an award, the consultant will also be rewarded. Consultants who do not come to the field will not be able to benefit from the mentioned award.

3.2.38. One person within the team, appointed as “**CAPTAIN**”, will carry out all communication and coordination processes with the TEKNOFEST Rocket Competition Committee.

3.2.39. During the competition, information will be provided by the TEKNOFEST Rocket Competition Committee only to the team captains.

3.2.40. The responsibility of the team captains for following the processes (*submitting an application, uploading the reports to the system before deadlines, filling out the forms, etc.*) will rest with the team captains, and the TEKNOFEST Rocket Competition Committee will not be responsible for any delay and/or disruption caused by them.

3.2.41. Applications must be made online through the www.t3kys.com application system before the deadline specified in the Competition Calendar.

3.2.42. On the application dates, the team captain must first be registered in the system, followed by the advisor and the other team members. After the team captain sends an invitation to the e-mail addresses of the advisor and the team members, the people to whom an invitation was sent log in to the Application System and accept the invitation from the "My Team Information" section. Thus, the registration is completed. Otherwise, the registration will not be completed.

3.2.43. All processes under the TEKNOFEST Rocket Competition (*submitting applications, submitting reports, submitting applications for travel financial support, submitting objections, adding/removing members, etc.*) are carried out via the Corporate Management System.

3.2.44. Within the scope of TEKNOFEST Rocket Competition, all processes (*Application, Report Retrieval, Travel Financial Support Application, Objection Processes, Member Addition/Removal Processes, etc.*) must be carried out by the team captains in coordination with the team advisor (*the signature of the team advisor indicating their approval must be included on all relevant documents*).

3.2.45. Members can be added and removed until the CDR delivery date.

3.2.46. Before applying, the contestant must read and approve all the provided details related to the competition, the conditions of participation (*the application will be considered as an indication that the contestant approves the rules*), and participation in the competition.

3.2.47. The first three teams in this category that meet the necessary conditions will be awarded cash prizes. The conditions for winning the prize and the prize amounts are explained in details in the relevant sections.

3.2.48. The TEKNOFEST Rocket Competition Committee has the authority to limit the number of team members to take place in the Aksaray Hisar Firing Test Range (*the test range where the competition finals are to be held*) for any reason. In the event of any restrictions, the necessary information will be provided by the TEKNOFEST Rocket Competition Committee.

3.2.49. Transportation and accommodation support to be provided to the finalist teams is limited. The number of people to be supported will be notified to the teams by the TEKNOFEST Rocket Competition Committee.

3.2.50. The TEKNOFEST Rocket Competition Committee is not responsible in any way for any damage caused by the teams to third parties throughout the competition.

3.2.51. The TEKNOFEST Rocket Competition Committee is not responsible in any way for any damage that the contestants suffer which are not caused by the TEKNOFEST Competition Committee throughout the competition.

3.2.52. The teams are responsible for designing, developing, producing, and testing their systems in accordance with the laws of their homeland and Republic of Türkiye. TEKNOFEST Rocket Competition Committee has no responsibility in the event of any violation in this regard.

3.2.53. TEKNOFEST Rocket Competition Committee reserves the right to make changes to those specifications in any time throughout the competition.

3.2.54. The applications of teams that do not meet the above conditions shall not be considered valid.

3.3. REQUIREMENTS FOR A4 INTERNATIONAL CATEGORY

This section contains the essential requirements for the rockets to be developed by the teams competing in the A4 competition category.

3.3.1.INDISPENSABLE REQUIREMENTS

3.3.1.1 Teams that fail to meet any of the indispensable common requirements will be eliminated.

3.3.1.2 The rocket design must be done in accordance with the Concept of Operations (CONOPS) of the A4 category.

3.3.1.3 For a rocket to be considered fully successful, it must perform a flight in accordance with the Operational Concept (CONOPS) of the A4 category (*the flight must fully satisfy the*

operational concept).

3.3.1.4 The teams are responsible for recovering all components and the payload of the rocket in a reusable condition following the launch.

3.3.1.5 All components and payload of the rocket must be recovered independently and in such a manner as to maintain their integrity.

3.3.1.6 Teams must ensure recovery by means of a parachute.

3.3.1.7 The payloads and the primary parachutes must separate within a maximum of two (2) seconds after the rocket passes apogee and enters the nose-down orientation of its flight trajectory.

3.3.1.8 The secondary parachutes responsible for recovering the rocket and its components must deploy when the rocket is between 400 and 600 meters above the ground.

3.3.1.9 It is mandatory that the rocket's current location data is shared uninterruptedly with the team's ground station via the communication computers on the system.

3.3.1.10 The rocket motors to be used by all finalist teams at the Aksaray Hisar Firing Range are commercially available solid-propellant products (*Aerotech/USA motors*) to be supplied and distributed by the TEKNOFEST Rocket Competition Committee, and the teams are not permitted to use any motor other than those.

3.3.1.11 The motors to be supplied by the Competition Committee will be handed over to each team at the Aksaray Hisar Firing Range.

3.3.1.12 Each team competing in the A4 International categories will be provided with one (1) Aerotech M1850 motor, and teams in this category must design their rockets by considering the motor to be delivered to them.

3.3.1.13 Rockets are to be recovered with two parachutes (*in Figure 3.1, the yellow parachute is the "Primary Parachute" and the green parachute is the "Secondary Parachute"*), while the payload must be recovered with a different parachute to that of the rocket (*the orange parachute in Figure 3.1*).

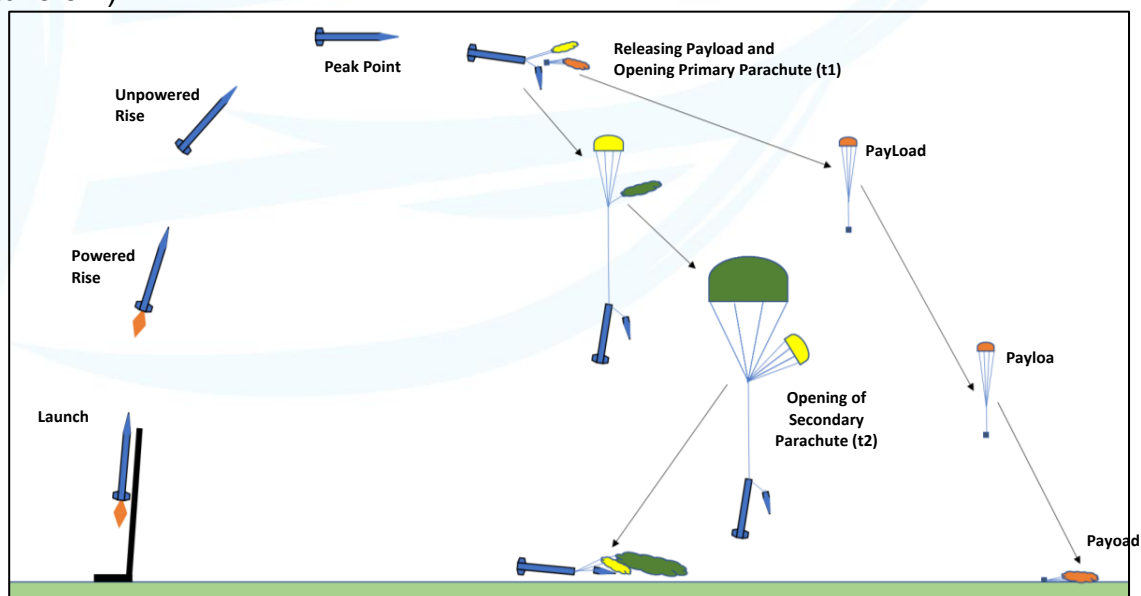


Figure 3.1 Operation Concept for A4 Category

3.3.1.14 The Earth and Rocket/Missile Coordinate Systems applicable in the TEKNOFEST

Rocket Competition are presented below. Details of Rocket/Missile & Earth Coordinate Systems and transformation methods for coordinate systems are given in Attachment-1;

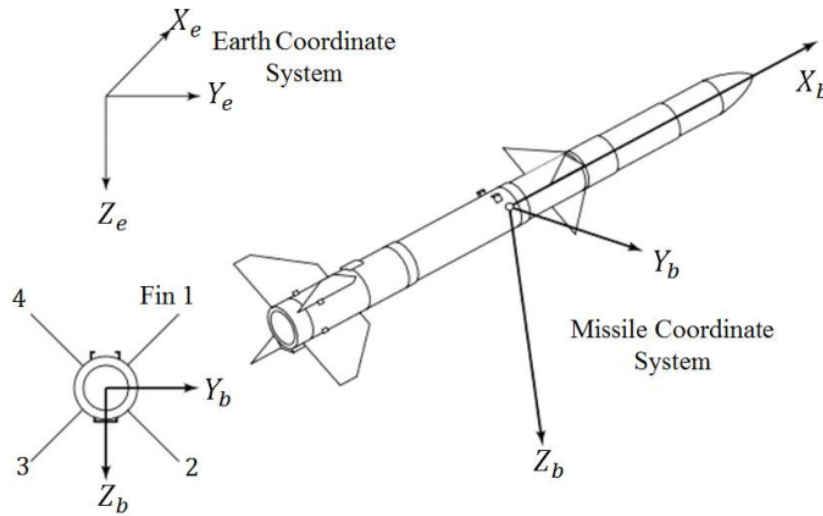


Figure 3.2 Earth and Rocket/Missile Coordinate System

3.3.1.15 Technical information about the motor to be provided to the teams for the TEKNOFEST-2025 Rocket Competition can be accessed from official websites of Aerotech/USA and TEKNOFEST Rocket Competition

(<https://www.teknofest.org/en/yarismalar/rocket-competition/>)

3.3.1.16 The rocket motors to be provided by the Competition Committee will be delivered to the teams that receive the necessary approvals from the referees during the assembly/integration activities of the finals at the Aksaray Hisar Firing Range. The motors shall be ready for assembly/integration with the rocket.

3.3.1.17 The motors should be installed on the rockets after all other installation and integration activities related to the rocket have been successfully completed and approval has been received from the referees (*at the final stage of the installation/integration activities*).

3.3.1.18 Teams are strictly prohibited from designing, manufacturing, and integrating into the rocket any components that will affect the performance of the motors (*increasing or decreasing thrust, directing thrust, etc.*).

3.3.1.19 Separate tracking systems (*i.e. GPS or radio transmitters*) must be provided for both the payload to be recovered and the rocket itself.

3.3.1.20 To record the maximum altitude to which the rockets fly, the teams will be provided with a referee altimeter (*technical details of referee altimeter are provided in Attachment-2*), which will be handed over to the teams that receive approval from the referees on the integration/assembly day at the Aksaray Hisar Firing Range by the TEKNOFEST Rocket Competition Committee (*Charging of the altimeter and ensuring that it remains operational until the completion of the launch are entirely the responsibility of the teams*).

3.3.1.21 It is the responsibility of the teams to charge their altimeter devices and keep them in working order until the launch is completed. Teams that fail to do this will not be considered in the competition.

3.3.1.22 Teams that complete the recovery operation must deliver the altimeters along with the recovered components of the rocket to the Competition Committee for evaluation. The

altitude data must be readable without any additional action. Any failure in this regard will result in the team's altitude not being considered by the TEKNOFEST Rocket Competition Committee.

3.3.1.23 Commendation awards will be given to teams in the categories of “BEST ROCKET,” “EXEMPLARY TEAM,” and “BEST ORIGINAL DESIGN” and details regarding these awards are provided in the relevant sections

3.3.1.24 “BEST ROCKET” Award:

The rockets brought to the Aksaray Hisar Launch Site will be evaluated in terms of manufacturing quality (including external and internal components and stages), paint/coating, lettering and symbols on the rocket, and the visuals on the rocket, from the perspectives of aesthetics and creativity. The most aesthetically pleasing rocket in each category will be awarded a plaque. This award is honorary in nature and does not carry any monetary value.

3.3.1.25 “EXEMPLARY TEAM” Award:

It is of great importance that teams are well prepared in advance for all their duties. Teams that demonstrate behavior in line with carefully prepared work plans; carry out all on-site tasks and responsibilities safely, effectively, and efficiently, and complete them on time (regardless of whether they obtain a launch attempt); manage crises calmly and proactively; reflect strong team spirit and energy at the site; set an example for other teams; support other teams at the site without violating competition rules (teams may share surplus materials with other teams with the approval of the Competition Committee); provide morale support to other teams; and strictly comply with Occupational Safety rules will be evaluated by the Competition Committee. This award is honorary in nature and carries no monetary value.

3.3.1.26 “BEST ORIGINAL DESIGN” Award:

It is important for the teams to use originally developed components and/or subsystems in their rockets. Teams are expected to present detailed information about the components and/or subsystems they have developed originally to the Competition Committee in the CDR and FRR (including design, manufacturing, integration, ground test results, etc.). This award is honorary in nature and carries no monetary value.

3.3.1.27 The mission performance table according to the recovery results of the rocket components and payload indicated by the TEKNOFEST Rocket Competition Committee is presented below;

COMPONENTS	RECOVERY PERFORMANCE		
Rocket Components	Successful	Partially Successful	Successful
Payload	Successful	Successful	Partially Successful
MISSION RESULT	COMPLETE SUCCESS	PARTIAL SUCCESS	PARTIAL SUCCESS

Table 3.7 Mission Performance Table

3.3.1.28 For any recovery success other than that listed in **Table 3.7** (if any of the Rocket Components or Payload recovery fails), the mission will be considered to have FAILED.

3.3.1.29 It is mandatory for the teams to perform trajectory simulations in accordance with the “Open Rocket” simulation menu in **Figure 3.2**, and to attach the trajectory simulation output created with Open Rocket to the relevant report. Any failure in this regard will lead to the non-evaluation of the report.

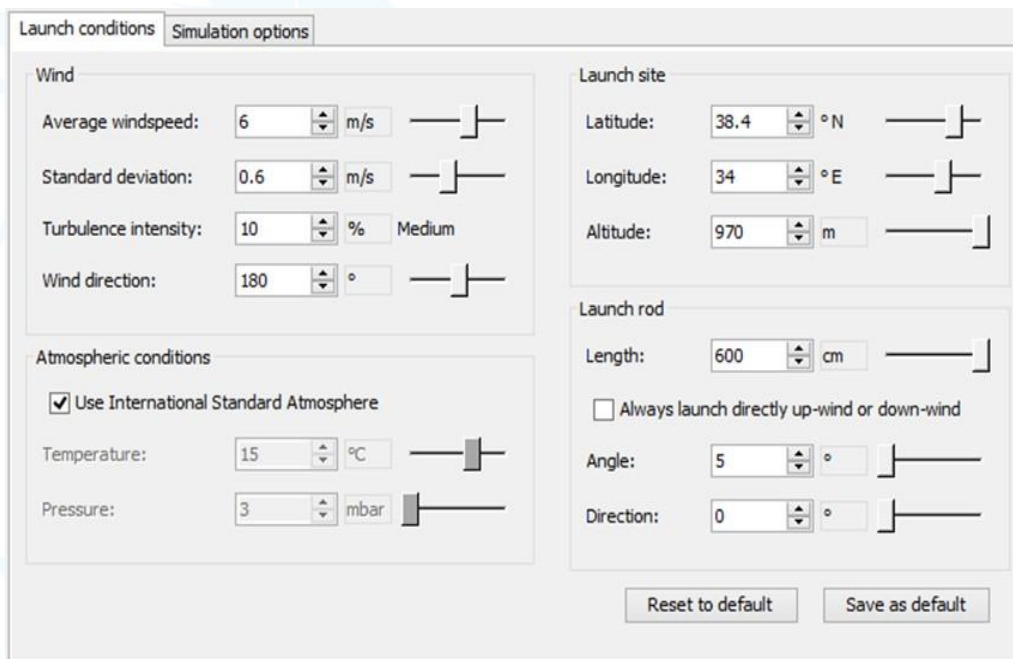
3.3.1.30 Trajectory simulation outputs generated using Open Rocket must be included in the

CDR and CDR reports.

3.3.1.31 The data in the teams' OpenRocket files must be identical to the Open Rocket data presented in the PDR and CDR reports.

3.3.1.32 Teams shall not enter their Mission Payload as "Unspecified Mass". The payload will be named "PAYLOAD", and its mass will be entered as 4,000 grams (4 kg) minimum.

3.3.1.33 The values on the "Launch Simulation" screen shown in **Figure 3.2** must be entered into the simulation. Reports of teams that have not performed a simulation with these values will not be evaluated.



The screenshot shows the 'Launch Simulation' window with two tabs: 'Launch conditions' and 'Simulation options'. The 'Launch conditions' tab is active and contains the following settings:

- Wind:**
 - Average windspeed: 6 m/s
 - Standard deviation: 0.6 m/s
 - Turbulence intensity: 10 % Medium
 - Wind direction: 180 °
- Atmospheric conditions:**
 - ☒ Use International Standard Atmosphere
 - Temperature: 15 °C
 - Pressure: 3 mbar
- Launch site:**
 - Latitude: 38.4 ° N
 - Longitude: 34 ° E
 - Altitude: 970 m
- Launch rod:**
 - Length: 600 cm
 - ☐ Always launch directly up-wind or down-wind
 - Angle: 5 °
 - Direction: 0 °

At the bottom right, there are two buttons: 'Reset to default' and 'Save as default'.

Figure 3.2 Open Rocket Simulation (*Launch Conditions Screen*)

3.3.1.34 Rockets will be launched with an elevation angle of 85° from the ground and a firing direction angle determined by the TEKNOFEST Rocket Competition Committee based on the prevailing wind direction.

3.3.1.35 The specifications of the launch pad are provided in Attachment-3.

3.3.1.36 When painting and/or coating their rockets, teams must comply with the TEKNOFEST Rocket Competition standards; otherwise, a penalty of up to 100 points may be applied. The standards related to the rockets are listed below;

- The "Rocket Name" must be clearly visible (*proportional to the rocket body*) and printed in an appropriate color and font on the upper section of the rocket body.
- The "Team Number" and "Team Name" must be clearly visible (*proportional to the rocket body*) and printed in an appropriate color and font on the middle section of the rocket body.
- The "Contact Person Information: Full Name and Mobile Phone Number" must be clearly visible (*proportional to the rocket body*) and printed in an appropriate color and font on the lower section of the rocket body.
- All text must be permanent and not easily removable.

3.3.1.37 In addition to the requirements stated in the competition specification, the requirements included in the PDR, CDR, and LRR templates are also indispensable components of the competition specification, and teams must design their rockets in accordance

with these requirements.

3.3.1.38 The cable connections inside the rocket must comply with the requirements listed below; otherwise, a penalty of up to one hundred (100) points may be applied;

- They must be in a twisted structure (*with at least 10 twists per centimeter of cable*), and the twisting must be performed in pairs or triplets among the High, Low, and Ground lines.
- The thickness of the battery cables must be determined according to the system's total continuous and instantaneous current, and cables in the AWG16–28 range must be used.
- Connectors must be suitable for screw-mounted installation, and for all other connectors selected, care must be taken to ensure that they are secured after installation, considering the vibration during flight.
- After cabling is completed, the continuity of each line must be verified by performing circuit continuity resistance tests, with a maximum allowable resistance of 1 ohm.
- For connectors containing pins, push-and-pull tests (*by applying manual force*) must be performed.
- Adhesive chemicals such as epoxy or silicone must be applied to the back of the pins in connectors containing pins.
- If additions need to be made to cable bundles, these additions must not coincide with the twist points of the cabling and must be performed in a staggered manner so as not to create excessive thickness.

3.3.1.39 At the Aksaray Launch Site, team members responsible for assembly/integration as well as launch and recovery activities are required to carry out all operations in compliance with Occupational Safety rules. Team members (*including advisors*) who act contrary to these rules will be penalized with 50 penalty points for the first violation and 100 penalty points for the second violation. If the team member (*including advisors*) continues to violate Occupational Safety rules, the team will be disqualified from the Competition.

3.3.2.RECOVERY SYSTEM REQUIREMENTS

3.3.2.1 Teams that fail to meet any of the recovery system requirements will be disqualified.

3.3.2.2 It is mandatory to use a parachute as a recovery system.

3.3.2.3 Primary parachutes are only required to reduce the velocity of the rocket or stage to 9–20 m/s and to prevent tumbling.

3.3.2.4 For rockets and components that must be recovered by a secondary parachute, the design and manufacturing must ensure a landing velocity between 5–9 m/s in order to prevent damage.

3.3.2.5 The payload must be recovered separately with its own parachute and without any connection to the rocket's components (*without being connected to any point by any equipment such as a shock cord, etc.*).

3.3.2.6 For the parachute recovery system, hot gas generators that make use of chemical reactions for the separation of the relevant components from the rocket (*gunpowder, etc.*), or pneumatic, mechanical, cold gas, or other systems developed by the team (*which are not harmful to the environment, and the risks of which can be managed*) can be used.

3.3.2.7 Teams will be allowed to use their own pyrotechnics as hot gas generators.

3.3.2.8 A capsule-shaped pyrotechnic Hot Gas Generator (HGG) can also be provided to the teams by the TEKNOFEST Rocket Competition Committee upon request (*to be handed over at Aksaray Hisar Firing Test Range*) and information about the HGG is provided in Attachment-4.

3.3.2.9 Non-commercial pressure vessels (*pressurized tanks, tubes, etc.*) shall not be allowed in the parachute separation process for safety reasons.

3.3.2.10 After the teams receive all the tags, they will receive their HGGs from the referees and hang the yellow flag on their workstation (*to indicate their receipt of the HGG*). They shall integrate the HGG into their rockets under the referee's control, and any failure in this regard will result in a 25-point penalty.

3.3.2.11 In team-developed original systems, no team-caused accidents should occur, and the Safety Ignition Units (SIUs) delivered to the teams by the Competition Committee must not be unintentionally activated due to negligence, careless handling, or improper work practices.

3.3.2.12 The process of filling commercial pressure vessels that can be used in the competition must be carried out in the installation/integration area and under the supervision of the referee committee.

3.3.2.13 Each parachute must have different colour and they should be easily distinguishable at a distance from the naked eye.

3.3.2.14 Teams must prove the location of the payload and rocket components that need to be recovered to the referee with data downloaded to the team's ground station.

3.3.2.15 No tightly fitted rocket components or subsystems (*such as the nose cone, airframe, etc.*) may be used in the recovery processes. For the required pressure and forces, shear pins must be used during recovery, and the related calculations and analyses must be demonstrated in the PDR and CDR reports

3.3.3. PAYLOAD REQUIREMENTS

3.3.3.1 Teams that fail to meet any of the payload requirements will be disqualified.

3.3.3.2 The payload must have a mass of at least four (4) kilograms.

3.3.3.3 The mass measurement of the payload will be carried out by the Competition Committee in the Integration area of the Aksaray Hisar Firing Range. To ensure the easy measurement of the payload weight, it must be designed and manufactured in such a way that it can be easily separated from the rocket.

3.3.3.4 If the payload meets the scientific task(s) requirements (*definitions, requirements, and examples regarding scientific task loads are included in Attachment-5*), the team will be awarded up to an additional 150 points.

3.3.3.5 The additional points that can be awarded to the teams related to the scientific task load and the evaluation criteria are presented in **Table 3.7** below;

DESIGN COMPLEXITY	SIMPLE		INTERMEDIATE		COMPLEX	
PERFORMANCE	PARTIAL *	FULL	PARTIAL *	FULL	PARTIAL *	FULL
EXTRA POINTS	0–49	50	51–99	100	101–149	150

** When the performance of the Scientific Payload is decided as "PARTIAL" by the referees, the additional points will be offered to the Chief Referee based on the evaluation of the referee committee, and the final additional points will be decided upon by the Chief Referee.*

Table 3.7 Scientific Payload Additional Point Table

3.3.3.6 Regardless of the design category (*Simple, Intermediate or Complex*) of the Scientific Payload, no additional points will be given to teams whose missions are unsuccessful.

3.3.3.7 The scientific payload must be separated from the rocket at the apogee of the flight (*the task to be performed by the scientific payload can be performed at any interval between the beginning and the end of the flight*) and the data to be acquired through the scientific mission(s) must be transmitted to the team's ground station at a frequency of at least 5 Hz. If data download cannot be provided at this frequency, even if the mission is fully successful, the performance of the payload will be considered partial.

3.3.3.8 The payloads to be used for carrying out a scientific task(s) cannot contain living organisms, corrosive chemicals, or radioactive materials; they must not be harmful to the environment/living organisms, and must not have any operational risks that are uncontrollable.

3.3.3.9 All payloads must be colored orange to ensure easy detection in the field during recovery operations. Each payload must include an audible warning system (*a buzzer that can be easily heard from at least 100 meters*) and a visual warning system (*flashing light*). The payload must also clearly display the following information in large black lettering: Team ID, Team Name, and "Phone Number to Call When Found"

3.3.4. AERODYNAMIC REQUIREMENTS

3.3.4.1 Teams that fail to meet any of the aerodynamic requirements will be disqualified.

3.3.4.2 There must be no change in the rocket's external body diameter along its flight axis.

3.3.4.3 There should not be no any gap larger than 0.1 mm between the airframe and its covers. Otherwise, the referees will not deem the rocket fit for launch until sealing measures are taken.

3.3.4.4 The use of movable flight control surfaces (*movable fins in the tail area*) that will provide active flight control is prohibited in the rockets.

3.3.4.5 Rockets in the A4 category must have a stability value between 1.5 and 2.5 at Mach 0.3.

3.3.4.6 Apart from external sensors, antennas, and cameras used on the rockets, no components that have no special mission or function and that create protrusions in the cross-sectional area or disrupt structural and aerodynamic integrity may be used.

3.3.4.7 On the OpenRocket main design page, the stability value for 0.3 Mach is calculated, and teams are required to present the stability value of their rockets at 0.3 Mach in both the PDR and CDR reports.

3.3.4.8 The minimum exit speed of the rockets at the launch ramp in A4 category is given in **Table 3.8**;

CATEGORY	A4
Minimum Liftoff Speed	25 m/sec

Table 3.8 Minimum Ramp Exit Speed for Category A4

3.3.4.9 Teams must prove to the TEKNOFEST Rocket Competition Committee in the relevant reports (*PDR and CDR*) and during the integration activities at the Aksaray Hisar Firing Test Range, that they meet the criteria in **Table 3.8**.

3.3.5. STRUCTURAL INTEGRITY REQUIREMENTS

3.3.5.1 Teams that fail to meet any of the structural requirements will be disqualified.

3.3.5.2 To ensure the internal and external pressures of the rockets are balanced, there must be at least three (3) holes with a diameter of 3.0-4.5 mm on the rockets.

3.3.5.3 The first hole must be in the front area of the rocket (*between the rocket nose and the front area of the airframe*), the second one in the middle area (*the area where the avionics are located*), and the third one in the area between the rear portion of the airframe and the motor.

3.3.5.4 Rockets must be able to withstand the structural loads encountered both during flight and during transportation and placement on the launch rail, and the design's compliance with these requirements must be demonstrated in the PDR and CDR reports.

3.3.5.5 It is forbidden to use PVC, compressed paper/kraft, or PLA materials on the surfaces of rockets that are exposed to aerodynamic forces (*airframe, fins, nose*).

3.3.5.6 Rockets must be resistant to the loads acting on aerodynamic surfaces (*body tube, fins, nose cone*) as well as the structural loads affecting the connections of related subcomponents (*motor mount, centering rings, etc.*). Compliance with these requirements and the corresponding design justification must be demonstrated in the PDR and CDR reports.

3.3.5.7 It must be demonstrated in the PDR and CDR reports that the rocket fins are resistant to flutter and that the design and flutter analyses have been carried out accordingly.

3.3.5.8 It must be proven in the relevant reports (*PDR, CDR, and LRR*) through analyses and test results (*tensile, compression, and torsion tests*) that all sub-components of the rockets are resistant (rigid) to the loads that may be imposed on the structural (*fin, engine block, centering ring, etc.*) connection points.

3.3.5.9 The eyebolts to be used must be manufactured as a single piece and made of cast steel.

3.3.5.10 The design must ensure that bent eyebolts and alternative materials that cannot withstand the loads applied to the eyebolt are not permitted for use..

3.3.5.11 The part of the nose cone that will enter the other airframe must be at least one and a half (1.5) times the outer diameter of the airframe.

3.3.5.12 The integration blocks that will be integrated into the airframes must enter both of them by at least 75% (0.75) of the outer diameter of the airframe.

3.3.5.13 A sample nose cone base is provided in **Figure 3.3**, and sample integration frames are in **Figure 3.4**.



Figure 3.3 Nose Cone

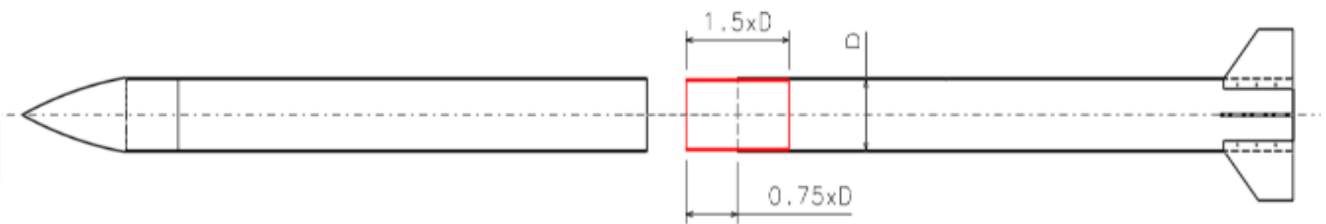


Figure 3.4 Integration Frame

3.3.5.14 The rail buttons will be hand-delivered to the teams by the TEKNOFEST Rocket Competition Committee after the payload is weighed during the integration activities in the Aksaray Hisar Firing Test Range.

3.3.5.15 A rocket must have at least two (2) rail buttons.

3.3.5.16 The rocket's center of gravity must be between the two rail buttons.

3.3.5.17 A technical drawing of the rail buttons to be handed over by the Competition Committee at Aksaray Hisar Firing Test Range can be found in Attachment-3.

3.3.5.18 The rail buttons must be attached to the airframe of the rocket where it is structurally strengthened.

3.3.5.19 The position of one of the rail buttons in the motor area (*between the center of gravity of the motor and the end of the fuselage*) is mandatory.

3.3.5.20 When attaching the rail buttons, it is mandatory to use fiber nuts, spring washers, and serrated washers, which create preload and prevent inadvertent loosening.

3.3.5.21 Parts protruding from the rocket's cross-sectional area (*sensors, antennas, and cameras*) must be attached to the rocket.

3.3.5.22 Parts that protrude into the rocket's cross-sectional area must be beyond the new center of mass of the rocket that emerges after the rocket has finished its burn.

3.3.5.23 It is mandatory for all switches related to the Flight Control Computer (FCC) and the payload to be at a maximum distance of 2.5 m from the rocket nozzle.

3.3.5.24 Activation of all switches on the rockets will be allowed when the rocket is on the launch ramp at a raised position for launching the rocket.

3.3.5.25 The design and production must not require any screw loosening/tightening or cover opening/closing while activating all keys.

3.3.5.26 The rocket must not suffer any structural deformation, the linearity of the rocket must be maintained, and the rocket airframe and nose cone connections must not open/loosen when lifted from the tip of the nose cone and the fins at the same time.

3.3.5.27 All lids on the rocket must be mechanically fastened and locked to prevent them from opening.

3.3.5.28 Fixing the covers on the rocket using methods that carry the risk of opening during flight (*due to the low pressures and vibrations experienced by the body during flight*) is prohibited.

3.3.6. AVIONICS REQUIREMENTS

3.3.6.1 Teams that fail to meet any of the Flight Control Computer requirements will be disqualified.

3.3.6.2 The rocket must contain two (2) Flight Control Computers (FCCs), one operating as the primary unit and the other as the backup.

3.3.6.3 It is mandatory for all recovery systems on the rocket to be managed by the rocket's primary and backup Flight Control Computers (FCCs).

3.3.6.4 The main FCC must be a COTS product (C-FCC) and the backup FCC must be an indigenously developed (I-FCC) one.

3.3.6.5 The commercial FCCs to be used by the teams must be selected from among the products listed in Attachment-6 (*products approved by the TEKNOFEST Rocket Competition Committee*).

3.3.6.6 The essential features that I-FCCs developed by the teams must possess, as well as the tests to which these I-FCCs will be subjected by the Competition Committee during assembly/integration activities at the Aksaray Hisar Firing Test Launch Site, are provided in Attachment-7. The I-FCCs must fully comply with all requirements specified in Attachment-7.

3.3.6.7 Starting from the moment the rocket is placed on the launch rail and continuing until the end of the flight, uninterrupted telemetry communication between the rocket and the ground station must be ensured. For this purpose, a communication system must be integrated into at least one of the Flight Control Computers or implemented as a separate system.

3.3.6.8 The communication system, which can be used to transfer telemetry data of the rockets to the team's ground station without interruption throughout the flight, can function independently or integrated with the FCC.

3.3.6.9 The flight algorithm in the original FCC must be indigenously designed by the team members.

3.3.6.10 It is mandatory for any of the FCCs to have a location determination and communication system.

3.3.6.11 If the I-FCC does not include a positioning and communication system, teams must use or develop an additional communication system.

3.3.6.12 Team member(s) who can technically answer the questions asked by the expert referees about the details of the indigenously designed flight algorithm must be present in the Aksaray Hisar Firing Test Range.

3.3.6.13 Teams whose I-FCCs make use of any COTS flight algorithms will be disqualified from the competition.

3.3.6.14 The team's I-FCC specialist must be competent enough to revise the flight algorithms in the assembly/integration area.

3.3.6.15 The I-FCC and C-FCC must operate completely independently of each other.

3.3.6.16 The I-FCC and C-FCC must have their own special processors, sensors, power supplies, and cabling.

3.3.6.17 The I-FCC and C-FCC must be connected to the separation system actuator with independent lines.

3.3.6.18 If either I-FCC or C-FCC fails partially or completely, the other must be able to fulfill the recovery functions of the rocket completely, without interruption.

3.3.6.19 There must be a minimum of two (2) different type of sensors connected to the I-FCC, while there is no minimum connection limit to the C-FCC (*sensors connected to different FCCs*).

can be the same).

3.3.6.20 At least two of the sensors connected to the I-FCC or C-FCC must be pressure sensors.

3.3.6.21 The embedded algorithm of the I-FCC must use data from at least three (3) different sensors

3.3.6.22 The sensors connected to different FCCs may be the same

3.3.6.23 At least two (2) independent criteria that will trigger separation sequences in the flight algorithms must be used.

3.3.6.24 Decision-making parameters must be based on the data read from the sensors.

3.3.6.25 The data read from the sensors must not be used directly, and any incorrect reading or sensor error must be taken into account.

3.3.6.26 Precautions to be taken for erroneous data from sensors (*filtering, etc.*) should be explained in details in the relevant design reports (*PDR and CDR*).

3.3.6.27 Teams that do not activate their recovery system using digital signals must use an Intermediate Electronic Component (IEC) with a digital ignition output in their system to activate the actuator.

3.3.6.28 The IEC must only evaluate the signals received from the FCC and must not perform any status assessment using sensor data.

3.3.6.29 If the recovery system is single, it must be controlled by both the main and the backup FCCs.

3.3.6.30 Redundancy of recovery system actuators is not mandatory.

3.3.6.31 There cannot be any electrical or wireless connection between the FCCs in the system.

3.3.6.32 The recovery trigger in the FCC algorithm must not be activated based on GPS or any type of counter data.

3.3.6.33 The actuators connected to recovery systems do not have to be redundant (*spring in a spring system, DC motor in a DC motor system, etc.*).

3.3.6.34 Recovery systems should not be activated unintentionally and in an uncontrolled manner.

3.3.6.35 All teams are required to have a ground station that can receive instant and continuous data from their rockets and payloads.

3.3.6.36 A ground station capable of continuously receiving data from the rocket and the payload is required.

3.3.6.37 To enable judges to simultaneously and live monitor the data displayed at the team's ground station, a separate computer screen connected to the team's ground station must be provided for the judging panel's use.

3.3.6.38 On launch day, the teams have a maximum of two (2) minutes to communicate with the ground station after the rocket avionics are activated.

3.3.6.39 If a team is unable to establish a healthy connection with the ground station at the end of the time allowed, the Competition Committee may grant the team the right to fly outside

the competition upon request (*a risk assessment will be carried out by the Competition Committee with the final decision made by the main responsible of the competition*).

3.3.6.40 Considering that rocket components may fall far from the ground station, the range of the transmitter and receiver antennas must be selected in accordance with the rocket's flight trajectory. The related analyses must be presented in the PDR and CDR reports.

3.3.6.41 The link bandwidth must be budgeted by assessing the power of the RF module, and this must be presented in the relevant design reports.

3.3.6.42 The avionics and sensors on the rocket must be able to operate easily under such conditions as vibration, pressure, and shock during the flight.

3.3.6.43 Teams must take the necessary protective measures against effects such as vibration, pressure, and shock, and must plan, conduct, and present the results of design verification tests in the CDR and LRR reports.

3.3.6.44 A mechanical key switch must be used as the on/off switch between the system's power sources (*battery, accumulator, supercapacitor, etc.*) and the circuits.

3.3.6.45 When the power connection is cut using the mechanical key switch, the power supply element must have no connection with any system component (*including indicators, power converters, and regulators*) and must not deliver power to any part of the system.

3.3.6.46 When a mechanical key switch is used to supply or cut power to the FCCs and the payload, a warning light or audible alert must be activated, and the switch must be embedded into the airframe considering aerodynamic effects. The necessary design and manufacturing steps must be carried out accordingly.

3.3.6.47 The energizing of the FCCs using such tools as ropes, shunts, screwdrivers, etc. is forbidden.

3.3.6.48 Turning OFF the mechanical switch should prevent power from being fed to any system component (*including LED indicators, power converters, and regulators*).

3.3.6.49 Teams that use Li-Po or similar batteries in the system must use a "Li-Po Safe Bag".

3.3.6.50 The team shall be responsible for the safety of the battery.

3.3.6.51 The FCCs on the rocket should be checked by turning on the switches while the rocket is on the launch rail.

3.3.7. SAFETY REQUIREMENTS FOR DESIGN AND PRODUCTION STAGES

3.3.7.1 Teams that fail to meet any of the design and manufacturing safety requirements will be disqualified.

3.3.7.2 The materials, equipment, and processes used in design and manufacturing stages must not cause any harm to human health or the environment.

3.3.7.3 Necessary security risk analyses and studies to reduce risks must be carried out within the scope of the design, production, and testing processes. It must be proven in the relevant design (*PDR, CDR, and LRR*) reports that these studies have been carried out.

3.3.7.4 Elements that may contribute to security risks in the design, production, integration, and firing activities must be determined in advance, the necessary precautions must be fully planned and proven to be implemented.

3.3.8. COMPETITION AREA REQUIREMENTS

3.3.8.1 Teams can take place at the Aksaray Hisar Firing Test Range for a maximum of three (3) days. The first two days are set aside for integration activities, while the third day is the dedicated to launch and recovery activities.

3.3.8.2 The activities to be carried out by the teams at the Aksaray Hisar Firing Test Range and the rules to be followed while in the field are presented in detail in Attachment-8.

3.3.8.3 For the A4 category, a maximum preparation time of 10 (ten) minutes will be allowed on the launch pad.

3.3.8.4 Competitors in teams must bring steel-toed work shoes when coming to the Aksaray Hisar Firing Test Range and wear them during the activities in this range. Competitors who do not comply with this rule will not be allowed into the range.

3.3.8.5 The teams are responsible for the safe delivery of their rockets to the Aksaray Hisar Firing Test Range.

3.3.8.6 The competing teams will be given a standard eight (8) hour period per day to carry out their integration activities. The Competition Committee has the authority to extend this period considering factors affecting the activities during the day.

3.3.8.7 Teams that cannot complete their integration activities on time and/or that fail to receive approval from the referees following inspections will not be allowed to launch as part of the competition (*it is strongly recommended that the rockets have a modular design and that the subsystems can be quickly integrated within 8-hour period*).

3.3.8.8 At the end of the integration day, the motors of the teams that pass the conformity checks by the referees will be installed on their rockets, which will be stored in a completely passive state (*when delivered to the Competition Committee the avionics systems must be off*) in the warehouse in the site by the Competition Committee until the rocket to be launched the next day.

3.3.8.9 Teams are allowed to use rotary tools and electric screwdrivers in integration operations at the Aksaray Hisar Firing Test Range.

3.3.8.10 Teams are forbidden from performing dangerous operations involving the attachment of disk cutting tips, etc. to dremels or cordless screwdrivers during integration activities at the Aksaray Hisar Firing Test Range (*because it is contrary to Occupational Safety*).

3.3.8.11 The use of the tools and equipment listed below by competitors in the integration area is forbidden. If any of the listed tools/equipment are found in this area during checks, they will be collected from the teams by the Competition Committee and returned to the team upon the departure of the team from the area (*It is important that teams come to the competition area ready, meaning that such equipment will not be needed*).

- Grinders, such as grinding motors, angle grinders, etc.
- Drills, such as drills connecting directly to a power source, etc.
- Milling cutter, hand milling machine, etc.
- Soldering equipment, such as soldering irons, etc.
- Staple and nail guns
- Buzzsaws and jigsaws

3.3.8.12 If the teams need to access tools and equipment that are not allowed in the Aksaray

Hisar Firing Test Range, they will be able to benefit from the workshop set up in the range by the Competition Committee.

3.3.8.13 Teams that receive or attempt to receive support for production or integration activities from a source other than the workshop established on-site by the Competition Committee will be eliminated from the competition.

3.3.8.14 Penalty points will be deducted from teams that have to use the workshop set up by the Competition Committee in the integration area (*deducted points depending on the duration of use of the workshop*).

3.3.8.15 Penalty points are given as listed in **Table 3.9**.

WORKSHOP USE TIME	PENALTY POINTS
1–15 Minutes (Small Scale Jobs)	15
16–30 Minutes (Medium Scale Jobs)	30
31–45 Minutes (Large Scale Work)	45

Table 3.9 Penalty Points to be Given In Case of Workshop Use

3.3.8.16 The works requested by the teams from the workshop will be evaluated by the authorized referee(s), the scale of the work to be done (*small-medium-large scale works*) will be evaluated, and penalty points will be applied by the Competition Committee.

3.3.8.17 For works that exceed 45 (forty-five) minutes and/or that cannot be achieved in the workshop facilities, no support will be received from the workshop established by the Competition Committee in the integration area.

3.3.8.18 A team can receive support a maximum of four (4) times from the workshop established in the integration area by the Competition Committee.

3.3.8.19 The maximum total support time a team will receive from the workshop established in the installation/integration area by the Competition Committee cannot exceed 60 (sixty) minutes.

3.3.8.20 Teams will benefit from additional points if they complete their integration before the specified time and become eligible to launch.

3.3.8.21 The additional reward points that can be earned by the teams based on their total integration times are given in **Table 3.10**.

TOTAL INSTALLATION/INTEGRATION TIME	REWARD POINTS
Category A4	
< 180 minutes	100
181–240 minutes	75
241–300 minutes	50

Table 3.10 Additional Points

3.3.8.22 Situations that result in the elimination of teams in the area where integration activities are to be carried out include the following:

- Failure to meet a requirement stated in the competition specification regarding the assembly/integration process that ends with the phrase “...teams that fail to meet any of the requirements will be disqualified,” or failure to correct a deficiency/issue related to this requirement on site,

- Failure to comply with the guidance and instructions of the integration area referees,
- Bringing prohibited materials/hand tools into the integration area and/or working with hand tools that may create sparks,
- Receiving or attempting to receive production or integration support from a source other than the workshop established on-site by the Competition Committee,
- The presence of more team members in the integration area than the maximum number defined in the competition specifications,
- Failure to obtain a label from the referees within the specified period, indicating that the team is entitled to launch their rocket,
- Failure to deliver the rocket to the Competition Committee within the specified time after receiving the label from the referees indicating that it is eligible to launch,
- The rocket must be delivered to the Competition Committee with any power source and/or electrical components active,
- Determination by the Competition Committee that the rocket does not meet the minimum strength requirements while on the launch rail,
- Failure to comply with Occupational Safety rules (*all team members including the team advisor*),
- Behaviour by team member(s) (*including the team advisor*) that violates the ethical rules of the Rocket Competition.

3.3.8.23 Situations that result in the elimination of teams on the launch day include the following:

- Failure to meet a requirement stated in the competition specification regarding the launch process that ends with the phrase "...teams that fail to meet any of the requirements will be disqualified," or failure to correct a deficiency/issue related to this requirement on site,
- Failure to follow the directions and instructions of the referees who is responsible for launching rockets,
- Loss of the team card (*with green labels*) and the labels received from the referees in the integration area indicating that the rocket can be launched,
- Exceeding the maximum 10 (ten) minutes preparation time given to the teams,
- Failure of the team to communicate with the ground station within two (2) minutes of the activation of the FCC(s) on the ramp,
- Damage given to the rocket by the team that would prevent its launch within the time allowed during the preparation period in the launching area,
- Changes/repairs/corrections, etc. on the systems during the operations carried out within the time allowed for launching the rocket,
- Powering up the avionics system on the rocket or the payload avionics without the permission of the authorised referee,
- Activating the separation system unintentionally for any reason at any stage prior to launch,

- Compromising the structural integrity of the rocket during its transportation to the rail or during loading onto the rail (*e.g. separation of the nose or frame*),
- Failure to comply with Occupational Safety rules (*all team members including the team advisor*),
- Team member(s) exhibiting behaviours that violate the ethical rules of the Rocket Competition.

4. CRITERIAS AND EXPECTATIONS

4.1. COMMON REQUIREMENTS REGARDING REPORT CONTENTS

4.1.1. Teams that fail to meet any of the requirements related to report preparation will be disqualified.

4.1.2. Competing teams must not copy content from their own or other teams' current or past reports, nor may they collaborate with other teams to conduct joint design, manufacturing, or testing activities.

4.1.3. When using tables, visuals, equations, or similar content that the team did not produce themselves, the source document must be properly cited within the report.

4.1.4. The teams are required to use the American Psychological Association-APA reference style for their content references.

(reference <https://apastyle.apa.org/products/publication-manual-7th-edition>).

4.1.5. Teams must prepare and submit to the Competition Committee a checklist that includes a "Compliance Matrix" to be placed in the appendix of the PDR and CDR reports (*a sample checklist is provided in Attachment-9*).

4.1.6. The checklist shall list the compliance status of the competition specification requirements. The list must include the following:

- Competition specification requirement number,
- Competition specification requirement statement,
- Compliance status (*to be indicated as Fully Met or Not Met*),
- The PDR or CDR page number where the fulfillment of the requirement is explained.

4.1.7. Teams must not share other teams' reports or work, even with references.

4.2. REPORTING

During the evaluation process, the teams will be asked to submit reports describing the status of their designs. These reports are:

- Preliminary Design Report (PDR)
- Critical Design Report (CDR)
- Launch Readiness Report (LRR)

4.2.1. PRELIMINARY DESIGN REPORT (PDR)

4.2.1.1 Teams that fail to meet any of the requirements related to the Preliminary Design Report will be disqualified.

4.2.1.2 Teams must prepare their Preliminary Design Reports (PDRs) in scientific article format.

4.2.1.3 The PDR template can be accessed through the official TEKNOFEST Rocket

Competition website.

4.2.1.4 Unsolicited information in the report will not be evaluated.

4.2.1.5 Teams are required to submit all the information requested in the PDR completely and in the relevant sections, and if the information that should be in the relevant section of the report is not presented as required, it will not be evaluated (*When teams do not present what is required of them in the relevant section, the risk of elimination of the team arises if the referees cannot see the information and analyses they want to see in that section. It is the team's responsibility to fully comply with the report template*).

4.2.1.6 Teams are obliged to deliver the PDR to the Competition Committee observing the deadlines in the competition schedule.

4.2.1.7 Teams are obliged to prove that the basic designs they made in the PDR fully meet the requirements of the Competition Specifications.

4.2.1.8 Teams must submit Open Rocket files with the “.ork” extension along with the PDR.

4.2.1.9 The information and data in the Open Rocket file submitted as an appendix to the PDR must be identical and consistent with the information and data presented in the PDR report.

4.2.1.10 To prove that the technical requirements have been met, a Compliance Matrix must be created by the teams (*a sample matrix is given in ATTACHMENT-9*) should be submitted as part of the PDR report.

4.2.1.11 In the Preliminary Design phase, where conceptual-level design activities will be carried out, teams must conduct component/subsystem/system-level trade-off analyses, use various optimization methods, and present rational design choices along with their justifications for different considerations.

4.2.1.12 In the trade-off analyses conducted by the teams, the decision criteria must be classified as “essential” and “optional,” and the selections made must be analyzed using clear cause-and-effect reasoning.

4.2.1.13 The material selection criteria and the compatibility of these selections with the system are also required to be included in the report.

4.2.1.14 The teams are required to present the PDR with the general CAD design of the rocket they plan to produce, explaining their systems and design in a detailed manner.

4.2.1.15 Preliminary analyses (*effects such as pressure, temperature, etc.*) that will constitute the basis for the Hot Gas Generator must be submitted in the PDR.

4.2.1.16 Teams must prepare a checklist that includes a “Compliance Matrix,” to be placed in the appendix of the PDR and CDR reports, and submit it to the Competition Committee (*a sample checklist is provided in Attachment-9*).

4.2.1.17 The information, analyses and evaluations requested to be included in the PDR by the Competition Committee must be presented in accordance with grammar rules, and in a manner that can be easily understood and followed. Teams that fail to meet this requirement will be subject to a deduction of points of up to 10% (ten percent) of the relevant section of the report, if necessary.

4.2.1.18 Teams that qualify (*teams that pass the threshold score which is to be announced in PDR phase*) in the Final Evaluation List during the PDR phase will proceed to the Critical Design Report (CDR) preparation stage.

4.2.2. CRITICAL DESIGN REPORT (CDR)

4.2.2.1 Teams that fail to meet any of the requirements related to the Critical Design Report will be disqualified.

4.2.2.2 All teams participating in the TEKNOFEST Rocket Competition are required to prepare their Critical Design Reports (CDR) in a scientific article format.

4.2.2.3 The CDR format and template will be shared with the competitors via TEKNOFEST website.

4.2.2.4 As part of their work for CDR, the teams will be responsible for making a detailed design, freezing the design, creating design outputs, fully meeting the requirements and task performance criteria given in the specifications, carrying out Project Management activities indicating that they are ready to move on to the production, integration and ground testing stages, and presenting the necessary evidence regarding their work to the Competition Committee.

4.2.2.5 As part of the analyses carried out during the CDR phase, all matters related to the rocket, including structural, fluid dynamics, flight algorithm proficiency etc., must be analysed and the outputs must be presented in the CDR.

4.2.2.6 Thus, the selected materials, production methods, flight resistance of the rocket and its components and the suitability of the flight algorithm must be proven in the CDR.

4.2.2.7 Numerical analyses related to structural strength and aerodynamic effects for the rockets and their components must be performed and presented in the CDR report.

4.2.2.8 Modelling and simulation processes are iterative, and the stages of rocket design should be presented in the CDR along with cause-effect relationships.

4.2.2.9 Teams must submit detailed information in the CDR on where, how and with what materials the relevant rocket subsystems and components (*body, nose, electronic card, etc.*) will be produced.

4.2.2.10 The teams should create a system integration diagram and explain it in detail (*The integration of all systems should be explained in details and supporting visuals taken from the CAD program should be presented in the report, answering such questions as: "How is the nose connected to the fuselage?", "How is the parachute connected to the fuselage?", "How is the engine fixed inside the fuselage so that it can be removed again?"*).

4.2.2.11 Images prepared with a Computer Aided Design (CAD) program should be used in the relevant sections of the CDR.

4.2.2.12 Production and ground testing activities should be planned and presented with their schedules in the CDR (*The content of the plans should include detailed information such as which production operations will be carried out in which weeks, on which dates the components will be tested, etc.*).

4.2.2.13 The fulfilment of each criterion in the competition specifications in the design resulting from the CDR phase (*at component/subsystem/system levels*) must be proven to the Competition Committee by the teams with a Compliance Matrix (*the matrix used in the PDR phase will be used in the CDR*).

4.2.2.14 All electronic components on the system powered by batteries must be specified in the CDR, which must include switching circuit schematics.

4.2.2.15 The Open Rocket file (*file with *.ork extension*) must be submitted with CDR.

4.2.2.16 The information and data in the OpenRocket file submitted as an appendix to the CDR must be identical or consistent with the information and data presented in the Critical Design Report.

4.2.2.17 The appendix of the Critical Design Report must include drawings and technical illustrations showing and visually explaining the integration of the rocket.

4.2.2.18 Information that is not requested in the Critical Design Report template will not be considered in the evaluation.

4.2.2.19 The teams are required to present in full all the information requested in the CDR and in the relevant sections. If information to be included in a particular section of the report is missing, it will not be evaluated (*When teams do not present what is required of them in the relevant section, the risk of elimination of the team arises if the referees cannot see the information and analyses they want to see in that section. It is the team's responsibility to fully comply with the report template*).

4.2.2.20 The teams that qualify for the final and receive financial support based on the CDR evaluation results will be announced by the Competition Committee on the date specified on the Competition Schedule.

4.2.2.21 Teams must prepare a checklist that includes a "Compliance Matrix," to be placed in the appendix of the PDR and CDR reports, and submit it to the Competition Committee (*a sample checklist is provided in Attachment-9*).

4.2.2.22 The information, analyses and evaluations requested by the Competition Committee to be included in the CDR should be presented in accordance with grammar rules, and in a manner that can be easily understood and followed. Teams that fail to meet this requirement will be subject to a deduction of points of up to 10% (ten percent) of the relevant section of the report, if necessary.

4.2.2.23 Teams must create and submit test videos for the prototype they have developed.

4.2.2.24 Design details and analyses (*strength, thermal and fluid dynamics analyses, etc.*) must be presented in details in the sub-system design reports (e.g. *the fluid dynamics analysis must contain details of the solution network used, boundary conditions, convergence details, fluid properties, results and interpretation of the results. Similarly, detailed information and evaluations must be included for other analyses.*)

4.2.3. LAUNCH READINESS REPORT (LRR)

4.2.3.1 Teams that fail to meet any of the requirements related to the Launch Readiness Report will be disqualified.

4.2.3.2 After the rockets are launched at the Aksaray Hisar Firing Test Range, all analyses, simulations, and ground test results required to ensure that the rockets successfully leave the launch rail in accordance with the specification requirements and perform their flight missions (*including the payload*) in compliance with the operational concept defined for their category must be included in the FRR.

4.2.3.3 The LRR must include a list of the team's distribution of tasks in the launch site, the safety measures that the team must take in the site, and the checks that the team will perform in the integration and launching areas.

4.2.3.4 In the LRR phase, it is mandatory that the entire (100%) system design and production be completed, and the teams that are successful in the LRR phase will come to the Aksaray Hisar Firing Test Range for the sole purpose of integrating and flying all rocket components (*Aksaray Hisar Firing Test Range is not the component manufacturing site for the teams*).

4.2.3.5 It is also necessary to prove in this report by means of a video footage that the rocket, after integration has been completed, can be made ready for launch within 10 (ten) minutes.

4.2.3.6 Considering the total mass of the completed rocket, a flight simulation of the rocket should be performed (*using Open Rocket and/or flight simulation and modelling software developed by the teams*) and the ramp exit speed should be calculated.

4.2.3.7 It must be proven that the launch rail lift off speed values obtained by the teams from the flight simulation performed with actual rocket data (*using data of the completed rocket, such as mass, Thrust/Weight, etc.*) are compatible with the relevant criteria in the Competition specifications (*speed exceeding the minimum speed for lifting off from the launch rail*).

4.2.3.8 When comparing the launch rail exit velocity values presented in the CDR and those obtained during the FRR phase using actual rocket data (*such as the mass of the completed rocket and thrust-to-weight values*), it must be demonstrated that the two values differ by no more than $\pm 5\%$.

4.2.3.9 All simulation data regarding the proofs to be presented in the LRR must also be included in the LRR (*The simulated Open Rocket file will be delivered with the LRR.*)

5. EVALUATION PRINCIPLES OF THE COMPETITION

5.1. OVERALL SCORE

5.1.1. Within the scope of Rocket Competition, teams in the A4 competition category will be evaluated over a total of 1000 (one thousand) points;

STAGE	Category A4
Preliminary Design Report (PDR)	100
Critical Design Report (CDR)	200
Launch Readiness Report (LRR)	100
Assembly & Integration Performance	200
Flight Performance	200
Recovery Performance (<i>Rocket and Payload</i>)	200
TOTAL	1000

Table 5.1 Scoring Details of Teams Competing in the A4 Competition Category

5.2. ASSEMBLY & INTEGRATION EVALUATION

5.2.1. Teams will be evaluated in four (4) different categories: "Payload and Recovery", "Avionics", "Structural Integrity" and "Aerodynamics" during the Assembly & Integration operations in the Aksaray Hisar Firing Test Range.

5.2.2. After this evaluation, those who first receive four (4) "Green" labels and then the "Grant Airworthiness" label (*the label that will be given to those who receive the final green label*) will be entitled to launch their rocket within competition.

5.2.3. Teams that fail to obtain all "Green" and "Grant Airworthiness" labels will not be given permission to launch their rocket out of the competition.

5.2.4. In the evaluation to be made by the Competition Committee for the teams that cannot obtain all the "Green" and "Grant Airworthiness" labels, the relevant teams can be given permission to launch their rocket out of the competition, if there is no safety issue regarding the flight of the rocket and if requested by the team.

5.2.5. To launch their rocket out of competition, teams must submit a written application (*signed both by the Captain and Advisor*) to the referee committee.

5.2.6. By submitting a petition, the teams accept that they will not be subject to scoring and will not be included in the ranking, even if they successfully complete the tasks given in the specification, in the event of them flying out of the competition.

5.2.7. If, after a team is accepted for an out-of-competition flight based on their submitted petition, any condition compromising flight safety is later identified, the Rocket Competition Chief Official has the authority to deny permission for the flight..

5.2.8. Teams must obtain the green labels for Aerodynamic and Structural Integrity as a minimum condition to launch their rocket out of competition.

5.2.9. Assembly & Integration Performance scoring is presented in **Table 5.2**;

ASSEMBLY & INTEGRATION PERFORMANCE SCORING	SCORE
Payload & Recovery System	50
Avionics	50
Structural Integrity	50
Aerodynamics	50
TOTAL	200

Table 5.2 Assembly & Integration Performance Scoring

5.3. FLIGHT AND RECOVERY EVALUATION

5.3.1. Teams will be evaluated on the success of the flight, with a total score of 200 (two hundred).

5.3.2. Teams will be evaluated on the recovery of their rocket components and payloads, with a total score of 200 (two hundred) points.

5.3.3. The scoring breakdown for the recovery of rocket components and payloads is provided below;

RECOVERY OF COMPONENT		RECOVERY STATUS		
		FAILED RECOVERY *	PARTIALLY SUCCESSFUL RECOVERY **	FULLY SUCCESSFUL RECOVERY
ROCKET COMPONENTS	Nose	0	1-24	25
	Body	0	1-24	25
	Motor	0	1-24	25

	Fins	0	1-24	25
	TOTAL	0	4-100	
PAYLOAD		0	1-99	100
* If the recovery of any rocket component (nose cone, airframe, motor, or fins) is unsuccessful, the rocket's total recovery score will be zero (0). ** The Competition Committee will evaluate the recovered components by considering structural damage and whether the components can perform their functions again, and will assign an appropriate score accordingly.				

Table 5.3 Recovery Performance Scoring

5.3.4. The minimum altitude that the rockets must reach is presented in relevant chapter. Teams that reach an altitude lower than this will be considered to have made an unsuccessful flight.

5.3.5. The flights performed by the teams will be listed and the altitude of the team that reached the highest altitude will be accepted as the reference.

5.3.6. The target altitude will be the value submitted to the Competition Committee in the FRR.

5.3.7. One of the success criteria (*deviation coefficient*) of the flight which is reaching the pre-planned flight altitude will be calculated using the "Altitude Deviation Coefficient (x)" formula below:

$$\text{Altitude Deviation Coefficient } (x) = 1 - \frac{|\text{Target Altitude} - \text{Altitude Reached}|}{\text{Target Altitude}}$$

5.3.8. The Deviation Coefficient (x) will be calculated considering the values recorded by the teams' altimeter.

5.3.9. Those outside the $\pm 15\%$ tolerance will not be taken into consideration and their Altitude Deviation Coefficient (x) will be assumed to be zero (0).

5.3.10. Devices other than the altimeter provided to the teams by the Competition Committee will not be used in calculating the points related to the target altitude reached.

5.3.11. The second success criteria (*altitude coefficient*) of the flight will be measured by the highest altitude that can be reached in the relevant category and the highest altitude in that category. This success will be calculated using the following "Altitude Coefficient (y)" formula:

$$\text{Altitude Coefficient } (y) = 1 - \frac{\text{Highest Altitude in the Team's Category} - \text{Altitude Reached by the Team}}{\text{Highest Altitude in the Team's Category}}$$

5.3.12. The teams' "Flight Performance Points" will be calculated using the following formula;

$$\text{Flight Performance Points} = 200 * x * y$$

5.3.13. When scoring and ranking the teams and determining the teams worthy of awards, the flight and recovery missions must achieve "Full Success" or "Partial Success".

5.3.14. In flight mission success evaluation, a flight will be deemed "Fully Successful" if the parachutes are deployed on time and completely (*the ropes and dome are not entangled when the parachute is first deployed, but they may be entangled later due to the wind*), and "Partially Successful" if the parachutes are deployed late and/or not fully deployed (*the ropes and dome are entangled when the parachute is first deployed*).

5.3.15. For the flight missions of the teams competing in the A4 competition category:

- In case of “**Full Success**”, the total score (100%) given by the Competition Committee to the team regarding assembly & integration performance, flight performance and recovery performance (*rocket and payload*) in the Aksaray Hisar Firing Test Range will be valid,
- In case of a “**Partial Success**”, half (50%) of the total points given by the Competition Committee to the team regarding assembly & integration performance, flight performance and recovery performance (*rocket and payload*) in the Aksaray Hisar Firing Test Range will be valid.

5.3.16. If the rocket and payload are recovered within a maximum of one (1) hour and presented to the Competition Committee, a recovery evaluation will be made; otherwise, the team will not be considered for recovery evaluation.

5.3.17. The evaluation of the recovery of the Rocket and the Payload (*Rocket and Payload will be evaluated separately out of 100 points*) will be evaluated by the referee panel and decided by the Chief Referee.

6. PRIZES

6.1. PRIZES FOR A4 CATEGORY

6.1.1. The table below shows the total amount that will be awarded to the teams that qualify for the prize. The first, second and third prizes shall be equally divided among all team members and deposited into the bank accounts specified by each member;

CATEGORY	FIRST	SECOND	THIRD
A4	5,000 USD	4,500 USD	4,000 USD

Table-6.1 TEKNOFEST-2025 Rocket Competition Awards

6.1.2. A prize payment of 300 USD (*three hundred American Dollars*) will be made to the advisors of the winning teams. If the advisor is not in attendance in the competition area, she or he will not be able to benefit from the prize in question.

6.2. MINIMUM SUCCESS CRITERIA FOR ELIGIBILITY

6.2.1. The flight and mission performances of the teams are divided into three: Full Success, Partial Success and Failure.

6.2.2. Teams failing to meet any of the following items will be considered “Failed”:

- The launched rocket must leave the launch rail without any issues,
- The rocket must fully complete its flight in accordance with the Operational Concept (CONOPS) specified for its category,
- Rockets must fly in accordance with the launch rail exit trajectory and reach apogee,
- The rocket must reach its target altitude within a $\pm 15\%$ tolerance,
- The payload must separate from the rocket immediately after apogee and successfully,
- The rocket components must be recovered intact as a whole (*nose cone, body, motor, and tail together*), and the payload must be recovered separately (*components and payload must land safely with parachutes and be reusable*),
- Teams that achieve a Successful or Partially Successful flight qualify for ranking in the award standings,
- Partially Successful teams cannot receive the first-place award, even if they rank

first by score at the end of the competition; they may receive, at most, the second-place award. Successful and Partially Successful teams are entitled to be included in the award ranking.

7. CODE OF CONDUCT AND OTHER RULES

7.1. CODE OF CONDUCT

7.1.1. Persons found to have exhibited any behaviour, actions, verbal expressions, etc. that are against social ethics in the festival area, or during the competition (*reporting stages, evaluation process, etc.*) will be disqualified from the competition, with legal action being taken against them immediately, and such persons will be banned from participating in any organisation or event hosted by the Turkish Technology Team Foundation (T3) for at least two (2) years.

7.1.2. Regarding the discourse and style used/to be used in all contacts and communications established with the Competition Committee:

- Rude and impolite (including insults) comments (verbal, written, etc.) and behaviours should be avoided,
- Insults, threats and bad language must be avoided;
- Competition officials and team members should not be targeted or subjected to accusations, whether directly or indirectly, through social media or any other communication tools, such as Facebook, Skype, Messenger, WhatsApp, X, LinkedIn, etc.
- Attention must be paid to spelling rules and style in all petitions and objections.

7.1.3. If a team and/or team advisor uses rude, discourteous, insulting, threatening, humiliating and accusatory expression(s) in any written and verbal communication with the Competition Committee, the team and its advisor will be eliminated from the competition altogether and, if necessary, they will be banned from participating in any organisation and event held by the Turkish Technology Team (T3) Foundation for a minimum of two (2) years.

7.1.4. The Competition Committee has the right to remove advisors and/or team members who do not comply with the code of conduct from the competition area when necessary.

7.1.5. Behaviours, acts, discourse, etc. that negatively affects the operation and motivation of other teams in the festival area are forbidden.

7.1.6. Legal action will be taken against any person(s) who act to the detriment of social harmony in the dormitory and in the proximity of the location where accommodation services are provided.

7.1.7. It is the responsibility of the teams to back up/store the equipment and materials to be used in the design and production processes, taking into account any potential problem that may occur, and to replace the parts when necessary.

7.1.8. Throughout the competition, it is forbidden to copy the reports from previous years or the current teams, or to engage in any collaborative designs, tests or analyses with other teams. Such acts are considered grounds for disqualification.

7.1.9. It is mandatory to behave objectively and in accordance with the requirements of the competition while in the competition area, without discrimination against language, religion, philosophical belief, political opinion, race, age or gender, and avoiding any behaviour or practices that hinder equal opportunities.

7.1.10. The goods/resources provided by a company/institution/organisation contributing to the competition to be used in the activities must be used exclusively in line with their purpose

and service requirements, and the said goods/resources should not be wasted. Wastefulness and extravagancy in buildings, vehicles and other public goods/resources must be avoided throughout the competition. It is of vital importance to behave effectively, efficiently and economically when using public goods, resources, workforces and facilities.

7.1.11. It is important to support activities that facilitate the conducting of the competition; to meet needs in the most effective, fast and efficient way; to increase service quality; and to ensure satisfaction throughout the course of the competition.

7.1.12. Competition area officials must abstain from behaviours that negatively affect, or that appear to affect, the conducting of their duties in an impartial and objective manner; and no request can be made that may benefit themselves/their relatives/friends or person(s) or organisations to which they are related. Legal action shall be taken against person(s) failing in this regard.

7.1.13. Team members will be held accountable for their responsibilities and liabilities, and must be open to transparent corporate evaluation and auditing; managers must take the necessary precautions to prevent corruption, as well as any actions and procedures that are in contravention with the objectives and policies of their institutions. They shall also train their staff in the principles of ethical behaviour, shall ensure their compliance with these principles and shall provide guidance in ethical behaviours.

7.1.14. While carrying out their tasks, team members must not make any statements, commitments, promises or suggestions that are binding on the institutions for which they work, and shall avoid making deceptive and false statements.

7.2. OTHER RULES

7.2.1. Objections fall under two categories: Report Objections and Final Stage Objections. Report Objections are received via the Corporate Management System within 48 (forty eight) hours after the report results are announced. The objections of teams that cannot submit their objection in accordance with the regulations at <https://teknofest.org/tr/competitions/objection/> within the period announced by the TEKNOFEST Competitions Coordination Office will not be accepted. The objections of the teams will be sent to different referee committees and re-evaluated.

7.2.2. Final Stage Objections can be made during and after the competition. Objections to be made after the competition must be made within 48 (forty eight) hours after the rankings are announced. Objections for any issue other than situations contrary to the specifications and violations of rules will not be accepted. No objections will be received for evaluations made based on quantitative criteria, and a new final or presentation evaluation will not be made. Such objections must be sent to iletisim@teknofest.org in the form of a petition, along with the reason for the objection. Objections made in accordance with the rules will be reviewed by the TEKNOFEST Competitions Coordination Office.

7.2.3. Teams will not post their objections on social media in any way. Otherwise, they will be eliminated from the competition.

7.2.4. Any work presented in the competition should not be seen as a product belonging to the team advisor, but as a product that was created with the collective effort of the competitor(s) in the team and the support of the advisor, reflecting the characteristics of all team members.

7.2.5. In the report, it should be stated on the relevant page that the teams that have benefited from the Previous Year Reports on the TEKNOFEST website have made citations. CITATION FORMAT: "Cited Phrase(s)" (Year, Competition Name, Category, Team Name)

7.2.6. The TEKNOFEST Rocket Competition Committee reserves the right to cancel the

competition in the event of there being an insufficient number of applications with the necessary technical knowledge and skills required for participation following the evaluations made as part of the application process.

7.2.7. The team captain, after registering on the system during the application period, is to ensure the full and correct registration of the advisor and team captain/members, in addition to sending them invites by e-mail. The invited member logs into the Application system to accept the invite, found in the “Team information” section, and the registration is duly completed. Otherwise, the registration is not done.

7.2.8. All processes required within the scope of the competition (Report Reception, Report Results, Financial Support Application, Objection Processes, etc.) It is done through the CMS portal. Teams should keep track of their processes via the CMS portal.

7.2.9. The competitors accept and undertake to participate in all kinds of written or visual promotions, publications, social media and Internet broadcasts related to the competition, whether made before or after the competition, by the T3 Foundation and/or TEKNOFEST. They further accept and undertake to make the relevant work available to the public within the framework of the T3 Foundation's open source policy.

7.2.10. I (We) declare that I (we) have used my (our) own ideas, knowledge and skills in selecting the project topic, that I (we) have received limited assistance from my (our) advisor and other relevant persons in relation to the problems that I (we) have encountered, that the said project belongs entirely to me (us), that TEKNOFEST has no obligation to ensure or monitor the protection of my (our) rights regarding the project, that TEKNOFEST cannot be held responsible for potential violations by third parties, that I (we) have not participated in another TEKNOFEST competition with the same project, and that I (we) have prepared the project in accordance with the rules specified in the competition specifications. I (We) agree and acknowledge that the project summary and the photographs submitted by me (us) and/or prepared by TEKNOFEST during the project application and/or competition process may be compiled and published by TEKNOFEST for various purposes, including but not limited to the promotion of the competition and compilation of statistics/archives.

7.2.11. Contestants may apply for registration or protection at the Turkish Patent and Trademark Office or WIPO (World Intellectual Property Organization) within the framework of the Industrial Property Law No. 6769 and other legal legislation regulating Intellectual Property Rights.

7.2.12. In the event of a competitor(s) violating the intellectual and industrial property rights of any product, the damages incurred by the TEKNOFEST Rocket Competition shall be compensated by the relevant team (including its advisor).

8. COMPETITION CALENDAR

The TEKNOFEST 2026 Rocket Competition schedule is presented in **Table 8.1**.

TEKNOFEST 2026 ROCKET COMPETITION SCHEDULE	
DATE	ACTIVITY
March 15 th , 2026	Application Deadline
March 17 th , 2026	Preliminary Design Report (PDR) Delivery Date
April 02-09, 2026	Announcement of Preliminary Design Report (PDR) Evaluation Results
June 1 th , 2026	Critical Design Report (CDR) Delivery Date
June 11-15, 2026	Announcement of Critical Design Report (CDR) Evaluation Results & Teams Eligible to Receive Financial Support Based on CDR Evaluations
June 16 th , 2026	Obtaining Undertaking and Bank Information from the Teams to Receive Financial Support Based on CDR Evaluations
July 22 st , 2026	Announcement of Finalist Teams Based on Launch Readiness Report (LRR) Evaluation
August 11-15, 2026	Announcement of Finalist Teams
August 20 th -25 th , 2026	Travel Financial Support Payments to be made to the Finalist Teams
September 07-13, 2026	Competition Finals at Aksaray Hisar Firing Test Range

Table 8.1 TEKNOFEST-2026 Rocket Competition Schedule

9TH YEAR

NATIONAL
TECHNOLOGY
INITIATIVE

